

SOLUTION

1. $v = an$

$$s = ut + \frac{1}{2}at^2$$

$$= a(n-2)2 + \frac{1}{2}a(n-2)^2$$

$$= \frac{na(n-2)}{2}$$

$$s = \frac{v(n-2)}{2}$$

2. $mgsin\alpha = mgsin\theta$

$$\alpha = \theta$$

3. $2Tsin\theta = W$

$$T = \frac{W}{2} \operatorname{cosec} \theta$$

$$(2) \frac{W}{2} \sec \theta$$

$$(3) W \cos \theta$$

$$(4) \frac{W}{3} \sin \theta$$

4. For platform

$$2F = N + 25g \quad \dots(1)$$

For person

$$F + N = 50g \quad \dots(2)$$

on solving 1 and 2

$$F = 250 \text{ N}$$

5. $kx = m_1g$

Just after BC is burnt

$$kx - m_1g = m_1a$$

$$a = 0$$

6. $3T \times v_0 = 2T \times v_A$

$$v_A = \frac{3v_0}{2}$$

$$v_{AB} = \frac{3v_0}{2} - v_0 = \frac{v_0}{2}$$

7. $\vec{a}_A + \vec{a}_B + \vec{a}_C + \vec{a}_D + \vec{a}_E = 3\hat{i}$

$$\vec{a}_B + \vec{a}_C + \vec{a}_D + \vec{a}_E = -\hat{i}$$

$$\Rightarrow \vec{a}_A = 4\hat{i}$$

$$\Rightarrow \vec{a}_B = 3\hat{i} - 24\hat{j}$$

$$\therefore |\vec{a}_A + \vec{a}_B| = 25 \text{ m/s}^2$$

8. $\mu g \sin \theta = \mu mg \cos \theta + \mu mg$

$$\sin \theta = \mu(\cos \theta + 1)$$

$$\sqrt{1 - \cos^2 \theta} = \mu(\cos \theta + 1)$$

Squaring on both side + solving

$$\theta = \cos^{-1} \left(\frac{1 - \mu^2}{1 + \mu^2} \right)$$

9. $w = fr$

$$= \mu NS$$

$$w = \mu mg \cos \theta \cdot S$$

10. $\frac{w_1}{w_2} = \frac{x_1}{x_2} = \frac{k_2}{k_1} (k_1 > k_2)$

$$w_1 < w_2$$

11. The speed is independent of initial direction.

12. Using C.O.M.E

$$0 + mg \times 1 = mg \times 0.5 + \frac{1}{2} \lambda v^2$$

$$v = \sqrt{10}$$

$$x = a \sqrt{\frac{2h}{g}}$$

$$= \sqrt{10} \sqrt{\frac{1}{10}} = 1$$

$$x = 1$$

13. $P = F.V.$

$$= 4500 \times 2$$

$$= 9 \text{ KW}$$

14. $F = -\frac{dv}{dx}$

15. $W_{PQ} + W_{QR} + W_{RP} = 0$

$5 + 2 + W_{RP} = 0$

$W_{RP} = -7$

$W_{PR} = 75$

16. $K \times 1 \times 10^{-2} = 2 \times 10$

$K = 2000$

Now,

$2000x = 3g$

$x = 1.5 \text{ cm}$

$\therefore 1.5 + 1 = 2.5 \text{ cm}$

17. $F = -\frac{dv}{dx}$

18. $w \propto v^2$

19. $X_{CM} = \frac{A_1 X_1 - A_2 X_2}{A_1 - A_2}$

$$= \frac{r \times 2r \times \frac{r}{2} - \frac{\lambda r^2}{2} \times \frac{4r}{3\lambda}}{r \times 2r - \frac{\lambda r^2}{2}}$$

$X_{cm} = \frac{2r}{3(4 - \pi)}$

20. $a = \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 g$

21. $K = \frac{P}{2\mu}$

$\mu = \frac{m_1 m_2}{m_1 + m_2}$

22. $MV' = MV$

$V' = \frac{MV}{m}$

$S = \frac{V'^2}{2a}$

$S = \frac{M^2 V'^2}{2m^2 \mu g}$

23. Using C.O.M.E

$\frac{1}{2} \frac{M_0 M}{(M + m)} (V_1 - V_2)^2 + 0 = \frac{1}{2} kx^2 + 0$

$x = (v_1 - v_2) \sqrt{\frac{mM}{(M + m)k}}$

24. $V_1 = \sqrt{2gh}$

$V_2 = \sqrt{\frac{gh}{2}}$

$m_1 v_1 = (m_1 + m_2) V_2$

$\Rightarrow m_1 = m_2$

27. Velocity along line of impact

$= \sqrt{2gh} \cos \theta$

$\therefore I = 2m \cos \theta \sqrt{2gh}$

28. $\bar{F} = \frac{d\bar{P}}{dt}$

29. $\Delta x = \frac{m(R - r)}{(M + m)}$

30. $I = \left[\frac{ml^2}{3} + m \left(\frac{\sqrt{5}}{2} l \right)^2 + \frac{ml^2}{12} \right] \times 2$

$I = \frac{10ml^2}{3}$

31. $F \cdot \Delta t = mV_{CM}$

$(F \cdot \Delta t) \frac{l}{2} = I_w = \frac{ml^2}{12} \cdot w$

$\Rightarrow V_{CM} = \frac{lw}{6}$

$\theta = wt$

$\frac{\pi}{2} = wt \Rightarrow t = \frac{\pi}{2\omega}$

$S = V_{CM} t = \frac{lw}{6} \times \frac{\pi}{2\omega}$

$$S = \frac{\pi l}{12}$$

$$\therefore x = \frac{l}{2} + \frac{\pi l}{12}$$

32. Only magnitude of $\vec{L}$ is constant

33. $v = \sqrt{4.5g\ell}$

34. $\frac{1}{2}kx^2 = mg \sin \theta \cdot x$

$$x = \frac{2mg \sin \theta}{k}$$

35. $\alpha = \frac{g}{8r}$

36. $v = rw$

$$3v = rw'$$

$$\frac{w'}{w} = 3$$

37. $ma \frac{l}{2} = mg \frac{b}{2}$

$$a = \frac{gb}{l}$$

38. $\int dF = \int_{\alpha}^{l+\alpha} \frac{Gm(a + bx^2)dx}{x^2}$

$$F = Gm \left(a \left(\frac{1}{\alpha} - \frac{1}{\alpha + l} \right) + bl \right)$$

39. $T = \frac{2\pi R}{V}$

$$\frac{mv^2}{R} = \frac{GMm}{R^{5/2}}$$

$$\Rightarrow T^2 \propto R^{T/2}$$

40. Gravitational field will be zero.

42. $\rho g(R - R \sin \theta) = \sigma g(R - R \cos \theta) + \sigma g R \sin \theta + \sigma g R \cos \theta$

$$\Rightarrow \tan \theta = \frac{\rho - \sigma}{\rho + \sigma}$$

$$\theta = \tan^{-1} \left(\frac{\rho - \sigma}{\rho + \sigma} \right)$$

43. $2\rho ghA - \rho ghA = \left(2\rho \frac{A}{2} \cdot \frac{l}{2} + \rho A \frac{l}{2} \right) a$

$$gh = 3 \frac{l}{2} a$$

$$h = \frac{3l}{2g} a$$

44. $p_0 + \rho gH - \rho 2aL = p_0$

$$a = \frac{gH}{2L}$$

45. $-\left(\frac{d^2h}{dt^2} \right) = \frac{ga^2}{A^2}$